

Light

MCQ

1. Light is a form of

- A) Matter
- B) Energy
- C) Force
- D) Sound

Answer: B

2. The sensation of sight is produced by

- A) Heat
- B) Sound
- C) Light
- D) Electricity

Answer: C

3. Objects that emit light are called

- A) Transparent objects
- B) Opaque objects
- C) Luminous objects
- D) Reflectors

Answer: C

4. The path followed by light is called

- A) Beam
- B) Ray
- C) Focus
- D) Axis

Answer: B

5. A group of parallel rays of light is called

- A) Image
- B) Reflection
- C) Beam

D) Shadow

Answer: C

6. Light always travels in a

A) Circular path

B) Zigzag path

C) Straight line

D) Curved path

Answer: C

7. Light is a type of

A) Mechanical wave

B) Sound wave

C) Electromagnetic radiation

D) Pressure wave

Answer: C

8. Tiny packets of energy present in light are called

A) Electrons

B) Protons

C) Neutrons

D) Photons

Answer: D

9. Which process uses light for food formation in plants?

A) Respiration

B) Transpiration

C) Photosynthesis

D) Germination

Answer: C

10. Which object reflects light without diffusion?

A) Rough surface

B) Mirror

C) Paper

D) Cloth

Answer: B

11. A mirror with a flat reflecting surface is called

- A) Convex mirror
- B) Concave mirror
- C) Plane mirror
- D) Parabolic mirror

Answer: C

12. Plane mirrors form images that are always

- A) Real
- B) Inverted
- C) Virtual
- D) Enlarged

Answer: C

13. Which image is formed when rays do not actually meet?

- A) Real image
- B) Virtual image
- C) Inverted image
- D) Enlarged image

Answer: B

14. Which mirror is commonly used as a shaving mirror?

- A) Plane mirror
- B) Convex mirror
- C) Concave mirror
- D) Parabolic mirror

Answer: C

15. A mirror with a curved reflecting surface is called

- A) Plane mirror
- B) Curved mirror
- C) Glass mirror
- D) Flat mirror

Answer: B

16. Which of the following is NOT a curved mirror?

- A) Concave

- B) Convex
- C) Parabolic
- D) Plane

Answer: D

17. A spherical mirror is a part of a

- A) Cone
- B) Cylinder
- C) Sphere
- D) Cube

Answer: C

18. Which mirror converges light rays?

- A) Plane mirror
- B) Convex mirror
- C) Concave mirror
- D) Glass slab

Answer: C

19. Which mirror diverges light rays?

- A) Concave mirror
- B) Convex mirror
- C) Plane mirror
- D) Parabolic mirror

Answer: B

20. Concave mirrors can form

- A) Only real images
- B) Only virtual images
- C) Both real and virtual images
- D) No images

Answer: C

21. Convex mirrors always form images that are

- A) Real and inverted
- B) Virtual and erect
- C) Real and enlarged

D) Virtual and inverted

Answer: B

22. Which mirror is used as a rear-view mirror in vehicles?

A) Plane mirror

B) Concave mirror

C) Convex mirror

D) Parabolic mirror

Answer: C

23. A parabolic mirror is a type of

A) Plane mirror

B) Convex mirror

C) Concave mirror

D) Glass slab

Answer: C

24. Solar cookers use

A) Plane mirrors

B) Convex mirrors

C) Parabolic mirrors

D) Glass sheets

Answer: C

25. The centre of the mirror is called

A) Focus

B) Pole

C) Axis

D) Radius

Answer: B

26. The centre of the sphere of which the mirror is a part is called

A) Pole

B) Focus

C) Centre of curvature

D) Radius

Answer: C

27. The distance between pole and focus is called

- A) Radius
- B) Diameter
- C) Focal length
- D) Axis

Answer: C

28. Focal length of a spherical mirror is equal to

- A) Radius
- B) Diameter
- C) Half the radius
- D) Twice the radius

Answer: C

29. The line passing through pole and centre of curvature is called

- A) Normal
- B) Principal axis
- C) Tangent
- D) Focus

Answer: B

30. A real image can be obtained on a

- A) Screen
- B) Mirror surface
- C) Eye lens
- D) Retina only

Answer: A

31. Virtual images cannot be obtained on a

- A) Mirror
- B) Eye
- C) Screen
- D) Glass

Answer: C

32. Laws of reflection apply to

- A) Only plane mirrors

- B) Only curved mirrors
- C) All reflecting surfaces
- D) Only glass

Answer: C

33. The angle of incidence is equal to

- A) Angle of refraction
- B) Angle of reflection
- C) Angle of deviation
- D) Angle of dispersion

Answer: B

34. The incident ray, reflected ray and normal lie in

- A) Different planes
- B) Same plane
- C) Parallel planes
- D) Vertical planes

Answer: B

35. Reflection from a smooth surface is called

- A) Diffused reflection
- B) Irregular reflection
- C) Regular reflection
- D) Random reflection

Answer: C

36. Reflection from a rough surface is called

- A) Regular reflection
- B) Specular reflection
- C) Irregular reflection
- D) Total reflection

Answer: C

37. Multiple reflection occurs due to

- A) Single mirror
- B) Rough surface
- C) More than one reflecting surface

D) Refraction

Answer: C

38. Number of images increases when angle between mirrors

A) Increases

B) Remains same

C) Decreases

D) Becomes zero

Answer: C

39. When two mirrors are parallel, images formed are

A) One

B) Two

C) Finite

D) Infinite

Answer: D

40. Kaleidoscope works on the principle of

A) Refraction

B) Dispersion

C) Multiple reflection

D) Scattering

Answer: C

41. Kaleidoscope is mainly used as a

A) Scientific instrument

B) Medical tool

C) Toy

D) Telescope

Answer: C

42. Periscope is used to

A) Measure distance

B) See microscopic objects

C) Observe over obstacles

D) Focus light

Answer: C

43. Periscope works based on

- A) Refraction
- B) Dispersion
- C) Laws of reflection
- D) Diffraction

Answer: C

44. Mirrors in a simple periscope are placed at

- A) 30°
- B) 45°
- C) 60°
- D) 90°

Answer: B

45. Periscopes are commonly used in

- A) Aeroplanes
- B) Submarines
- C) Cars
- D) Telescopes

Answer: B

46. Refraction of light occurs when light travels

- A) In the same medium
- B) In vacuum
- C) From one medium to another
- D) Along a mirror

Answer: C

47. Bending of light at the boundary of two media is called

- A) Reflection
- B) Dispersion
- C) Refraction
- D) Diffusion

Answer: C

48. Light bends towards the normal when it travels from

- A) Denser to rarer medium

- B) Rarer to denser medium
- C) Vacuum to vacuum
- D) Same medium

Answer: B

49. Light bends away from the normal when it travels from

- A) Rarer to denser medium
- B) Denser to rarer medium
- C) Air to air
- D) Water to glass

Answer: B

50. Pencil appears bent in water due to

- A) Reflection
- B) Refraction
- C) Dispersion
- D) Diffraction

Answer: B

51. Refractive index is the ratio of

- A) Angles
- B) Wavelengths
- C) Speeds of light
- D) Frequencies

Answer: C

52. Refractive index has

- A) Unit
- B) Dimension
- C) No unit
- D) Only SI unit

Answer: C

53. Refractive index of a medium is always

- A) Less than 1
- B) Equal to 1
- C) Greater than 1

D) Zero

Answer: C

54. The symbol for refractive index is

A) n

B) r

C) μ

D) f

Answer: C

55. Snell's law relates angles of

A) Reflection

B) Dispersion

C) Incidence and refraction

D) Deviation

Answer: C

56. The ratio $\sin i / \sin r$ is constant for

A) Reflection

B) Refraction

C) Diffusion

D) Scattering

Answer: B

57. Splitting of white light into colours is called

A) Reflection

B) Refraction

C) Dispersion

D) Scattering

Answer: C

58. The band of seven colours is called

A) Spectrum

B) Beam

C) Image

D) Ray

Answer: A

59. VIBGYOR represents

- A) Seven mirrors
- B) Seven rays
- C) Seven colours
- D) Seven images

Answer: C

60. Which colour has the shortest wavelength?

- A) Red
- B) Yellow
- C) Green
- D) Violet

Answer: D

61. Which colour deviates the most in a prism?

- A) Red
- B) Orange
- C) Blue
- D) Violet

Answer: D

62. Which colour deviates the least in a prism?

- A) Violet
- B) Blue
- C) Green
- D) Red

Answer: D

63. Dispersion occurs because different colours have different

- A) Mass
- B) Speed
- C) Density
- D) Shape

Answer: B

64. Rainbow is formed due to

- A) Reflection only

- B) Refraction only
- C) Dispersion of sunlight
- D) Scattering

Answer: C

65. Water droplets act like tiny

- A) Lenses
- B) Mirrors
- C) Prisms
- D) Screens

Answer: C

66. Rainbow is seen on the

- A) Same side of the Sun
- B) Opposite side of the Sun
- C) Above the Sun
- D) Below the Sun

Answer: B

67. Reflection of light from a plane mirror follows

- A) One law
- B) Two laws
- C) Three laws
- D) Four laws

Answer: C

68. The normal is a line drawn

- A) Along the surface
- B) Tangent to surface
- C) Perpendicular to surface
- D) Parallel to ray

Answer: C

69. The angle between incident ray and normal is called

- A) Angle of reflection
- B) Angle of refraction
- C) Angle of incidence

D) Angle of deviation

Answer: C

70. The angle between reflected ray and normal is called

A) Angle of incidence

B) Angle of reflection

C) Angle of refraction

D) Angle of dispersion

Answer: B

71. Concave mirrors are used in headlights because they

A) Scatter light

B) Diverge light

C) Converge light

D) Absorb light

Answer: C

72. Ophthalmoscope uses a

A) Convex mirror

B) Plane mirror

C) Concave mirror

D) Glass slab

Answer: C

73. Convex mirrors provide

A) Narrow field of view

B) No image

C) Wider field of view

D) Magnified image

Answer: C

74. Image formed by convex mirror is always

A) Real

B) Inverted

C) Virtual

D) Enlarged

Answer: C

75. The distance between mirrors in a periscope depends on

- A) Colour of light
- B) Purpose
- C) Size of mirror
- D) Angle only

Answer: B

76. Fibre optic periscopes are used by

- A) Engineers
- B) Teachers
- C) Doctors
- D) Pilots

Answer: C

77. Reflection from wavy water surface is an example of

- A) Regular reflection
- B) Total reflection
- C) Irregular reflection
- D) Refraction

Answer: C

78. Reflection from calm water surface is an example of

- A) Irregular reflection
- B) Diffused reflection
- C) Regular reflection
- D) Multiple reflection

Answer: C

79. A beam of light consists of

- A) Single ray
- B) Many rays
- C) Photons only
- D) Waves only

Answer: B

80. The speed of light is maximum in

- A) Glass

- B) Water
- C) Air
- D) Vacuum

Answer: D

81. Refractive index depends on

- A) Colour only
- B) Medium
- C) Shape of object
- D) Size of mirror

Answer: B

82. Which phenomenon explains bending of light in glass slab?

- A) Reflection
- B) Refraction
- C) Dispersion
- D) Diffusion

Answer: B

83. Which device does NOT use mirrors?

- A) Kaleidoscope
- B) Periscope
- C) Telescope
- D) Microscope

Answer: D

84. Reflection involves change in

- A) Medium
- B) Speed
- C) Direction
- D) Wavelength

Answer: C

85. Refraction involves change in

- A) Direction and speed
- B) Direction only
- C) Speed only

D) Frequency

Answer: A

86. White light is a mixture of

A) Two colours

B) Three colours

C) Five colours

D) Seven colours

Answer: D

87. The colour with highest wavelength is

A) Violet

B) Indigo

C) Green

D) Red

Answer: D

88. Parabolic mirrors are used to

A) Diffuse light

B) Focus energy

C) Scatter light

D) Absorb light

Answer: B

89. Reflecting telescopes use

A) Plane mirrors

B) Convex mirrors

C) Concave mirrors

D) Glass slabs

Answer: C

90. The reflected ray lies on the

A) Same side of normal as incident ray

B) Opposite side of normal

C) Same path as incident ray

D) Normal itself

Answer: B

91. Which mirror always gives diminished images?

- A) Plane mirror
- B) Concave mirror
- C) Convex mirror
- D) Parabolic mirror

Answer: C

92. Refraction does not occur when light travels

- A) From air to water
- B) From water to glass
- C) From air to air
- D) From glass to air

Answer: C

93. A prism is used to demonstrate

- A) Reflection
- B) Refraction
- C) Dispersion
- D) Diffraction

Answer: C

94. Formation of spectrum is due to

- A) Reflection only
- B) Different speeds of colours
- C) Same wavelength
- D) Scattering

Answer: B

95. The focal point of a convex mirror lies

- A) In front of mirror
- B) On the surface
- C) Behind the mirror
- D) At infinity only

Answer: C

96. The image formed by plane mirror is

- A) Laterally inverted

- B) Inverted vertically
- C) Diminished
- D) Enlarged

Answer: A

97. Which mirror is used in security surveillance?

- A) Plane mirror
- B) Concave mirror
- C) Convex mirror
- D) Parabolic mirror

Answer: C

98. Light energy helps in regulating

- A) Heart beat
- B) Sleep-wake cycle
- C) Digestion
- D) Hearing

Answer: B

99. Light that helps in electricity generation is

- A) Tube light
- B) Candle light
- C) Sunlight
- D) Bulb light

Answer: C

100. The study of behaviour of light is part of

- A) Chemistry
- B) Biology
- C) Physics
- D) Geology

Answer: C